ANIL R. TALEKAR

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ISTQB Certified | **HIL Engineer** | **ECU Verification & Validation** | **ADAS & Automotive Protocols** | **Automation with Python & Robot Framework**

CAREER OBJECTIVE:

To build a challenging career in the **Automotive and Electrical industry**, leveraging expertise in **HIL testing**, **ECU verification & validation**, and **automation with Python and Robot Framework**. Seeking to contribute to a progressive organization by applying technical skills, enhancing knowledge, and driving the growth of both the company and personal career development.

PROFESSIONAL SUMMARY

- **Embedded Test Engineer** with 8+ years of experience in the Automotive and Electrical industry, specializing in embedded systems validation and ECU testing.
- **HIL Testing expertise** using ETAS, OPAL-RT, dSPACE, and NI HIL; skilled in developing and executing automated test frameworks with Robot Framework and Python.
- Strong knowledge of **communication protocols** including CAN, CAN-FD, UDS, J1939, XCP, and CCP; proficient in creating Rest Bus simulations and CAN databases in CANoe.
- Experienced in **ADAS systems** validation and simulation; developed panels and simulations in CANoe to support advanced driver assistance testing.
- Skilled in **CAPL and Python scripting** for automation and test case execution; hands-on with Embedded C programming and debugging.
- Knowledgeable in **software development life cycle** (SDLC, STLC) and methodologies including Agile, V-Model, and Waterfall.
- Proficient with **tools** such as JIRA, RQM, DOORS, GIT, and Jenkins for test management, version control, and CI/CD.
- Experienced in **electronic test equipment** including function generators, oscilloscopes, multimeters, and power supplies.
- Recognized for strong communication skills, adaptability to new environments, and effective teamwork in cross-functional projects.

TECHINCALSKILLS

- **Communication Protocols**: UART, I2C, SPI, CAN, CANFD, UDS, J1939, CANTP, XCP/CCP
- **Programming Languages**: Python, CAPL, Embedded C
- **Frameworks**: Robot Framework, Pytest Framework
- **Tools**: CANoe, CANalyzer, CANape, NI Teststand, Trace32, DOORS, RQM, JIRA, ECU-Test
- **Version Control & CI/CD**: GIT, Jenkins

EDUCATIONDETAILS

- **PG Diploma in Advanced Embedded Systems** — CDAC, Mumbai *(2016)*
- **Bachelor of Engineering in Electronics** — TKIET College, Shivaji University, Kolhapur *(2015)*
- **Higher Secondary Certificate (HSC)** — RCSM Junior College, Kolhapur, Maharashtra State Board *(2011)*
- **Secondary School Certificate (SSC)** — Madhyamik Vidyalaya, Kasba Arale, Maharashtra State Board *(2009)*

WORK EXPERIENCE :

ACL Digital Pvt Ltd, Pune *Automotive & Electrical (eMobility)* **July 2024 – Present**
John Deere India Pvt Ltd, Pune *Automotive – Agricultural, Construction, and Forestry Equipment* **April 2022 – July 2024**
JCB India Pvt Ltd *Automotive – Agricultural, Construction, and Forestry Equipment* **November 2019 – April 2022**
Minda Stoneridge Pvt Ltd *Automotive Domain* **May 2019 – October 2019**
M-Tech Innovation Pvt Ltd *Automotive Domain* **September 2017 – April 2019**

PROJECT DETAILS:

Project #1: ADAS testing - RADAR ECU and BSW Testing
Tools: Vector CANoe, DOORS, JIRA, RQM, **Domain:** Automotive – Advanced Driver Assistance Systems (ADAS)
Description: Performed verification and validation of both Base Software (BSW) and Application Software (ASW) features on RADAR ECU. Scope included customer diagnostics, communication management, fault management, and ADAS functions such as Adaptive Cruise Control (ACC), Automatic Emergency Braking (AEB), and Lane Keeping Assist (LKA).
Roles & Responsibilities:

- **Requirements Review**: Analyzed and reviewed software requirements for RADAR features including Communication (COM), Customer Diagnostics (CD), and Fault Management (FM).
- **Fault Injection**: Triggered faults using CAN signals, environment variables, and XCP commands to validate ECU fault handling.
- **Test Case Development**: Authored manual test cases in RQM based on DOORS requirements; designed automated test cases using CAPL scripting.
- **Test Execution**: Executed test scripts on HIL test bench setups for ACC, AEB, LKA, COM, CD, and FM features.
- **Defect Management**: Logged and tracked failed test cases in JIRA; collaborated with developers to resolve defects.
- **Documentation**: Documented all test case results and shared reports with team management for review and sign-off.

Project #2: Advantor ECU Testing & Automation
Tools: NI HIL, VeriStand, TestStand, JIRA, RQM **Domain:** Automotive – ECU Functional & Diagnostic Testing
Description: Performed verification and automation of ECU features using NI HIL architecture, VeriStand, and TestStand. Scope included SPN/FMI diagnostics, broadcast message validation, sensor input automation, parameter writing, and address claiming.
Roles & Responsibilities:

- **Manual & Automated Testing**: Developed and executed manual and automated test cases for SPN639 (FMI14, FMI9), SPN33 (FMI05, 04, 03), and SPN191/161 (ISS & OSS sensors).
- **Broadcast Validation**: Conducted DM13 start/stop broadcast testing to ensure proper ECU communication.
- **Parameter Writing**: Validated ACN parameter writing using command linker.
- **Address Claiming**: Authored and executed address claiming test cases for ECU identification.
- **Message Validation**: Designed and tested EATON identification message cases.
- **Automation**: Implemented automation workflows using NI TestStand integrated with VeriStand for ECU regression testing.
- **Requirement & Defect Management**: Authored test cases in RQM, reviewed requirements, and tracked defects in JIRA.

- **Architecture Validation**: Validated NI HIL architecture setup for ECU functional and diagnostic testing.

Project #3: Primary Display Unit (PDU) and Cluster Testing for Tractor, Sprayer, and Harvester
Tools: CANalyzer, ETAS HIL, TractorSim, ECU-Test, Rally, DOORS **Domain:** Automotive – Instrument Cluster & Display Systems
Description: Validated PDU and cluster features including speedometer, fuel gauge, temperature gauge, and telltales (high beam, low beam, indicators, park lights) for agricultural vehicles such as tractors, sprayers, and harvesters. Focused on both manual and automated testing with Python and ECU-Test to ensure compliance with system requirements and functional accuracy.
Roles & Responsibilities:

- **Requirements Review**: Reviewed system requirements (QPL11) as part of Scrum team activities.
- **Test Case Development**: Created manual test cases based on DOORS system requirements; automated test cases using Python in Tracetronic ECU-Test tool.
- **Test Execution**: Executed tests on HIL (ETAS) and SIL (TractorSim) environments; maintained test reports on GitHub.
- **Review & Delivery**: Participated in QPL17 test result reviews prior to tag delivery; raised observations and defects during pre-delivery and post-delivery phases.
- **Defect Management**: Verified issues, collaborated with developers to resolve defects, and performed diagnostic testing.
- **Root Cause Analysis**: Identified issues/defects, conducted RCA, and proposed corrective solutions.
- **Collaboration**: Coordinated with developers, customer frontends, and cross-functional teams to discuss technical problems, report bugs, and support resolution.
- **Environment Maintenance**: Maintained HIL and SIL test environments in collaboration with respective teams.

Project #4: CAB Controller Testing (OSIO1 & OSIO2) for Tractor
Tools: CANalyzer, ETAS HIL, TractorSim, Rally, DOORS, ECU-Test **Domain:** Automotive – Tractor Control Systems
Description: Validated analog and digital switch functionalities in CAB Controller including brake pedal, acceleration/deceleration pedal, indicator switch, high/low beam switch, park switch, reverser (left/right hand), light switches, and power management systems for tractors. Focused on both manual and automated testing using Python and ECU-Test to ensure compliance with system requirements and functional accuracy.
Roles & Responsibilities:

- **Requirements Review**: Reviewed system requirements (QPL11) as part of Scrum team activities.
- **Test Case Development**: Created manual test cases based on DOORS requirements; automated test cases using Python in Tracetronic ECU-Test tool.
- **Test Execution**: Executed tests on HIL (ETAS) and SIL (TractorSim) environments; maintained test reports on GitHub.
- **Review & Delivery**: Participated in QPL17 test result reviews prior to tag delivery; raised observations and defects during pre-delivery and post-delivery phases.
- **Defect Management**: Verified issues, collaborated with developers to resolve defects, and performed diagnostic testing.
- **Root Cause Analysis**: Identified issues/defects, conducted RCA, and proposed corrective solutions.

- **Collaboration**: Coordinated with developers, customer frontends, and cross-functional teams to discuss technical problems, report bugs, and support resolution.
- **Environment Maintenance**: Maintained HIL and SIL test environments in collaboration with respective teams.

Project #5: Functional Testing of TCU Features in JCB Machines
Tools: CANoe, NI TestStand, DOORS, Bugzilla **Domain:** Automotive – Backhoe Loader & Excavator
Description: Validated TCU functionalities in JCB machines including Easy Shift Transmission, Smooth Ride System, Guide Me Home, and CNG ECU control for electrical, mechanical, and dual-fuel systems.
Roles & Responsibilities:

- **Bench & Black-Box Testing**: Performed open-loop bench and black-box validation for ECU features.
- **Test Case Development**: Authored and reviewed test cases, requirements, and code.
- **CAN Data Analysis**: Analyzed logs via JCB Service Master diagnostics tool.
- **Defect Management**: Reported, tracked, and resolved defects using Bugzilla; performed RCA in Embedded C.
- **Communication Testing**: Conducted CAN communication validation for ECU interactions.

Project #6: High Voltage Motor Control using Voltage by Frequency (V/Hz)
Tools: OPAL-RT HIL, JAMA, JIRA, Robot Framework, Python (PyCharm IDE) **Domain:** Electrical – Motor Control Systems
Description: Verified motor control features including variable frequency control, slip monitoring, torque boost, oscillation stabilization, and flux braking. Focused on validating performance and stability of high-voltage motors under different operating conditions with automated test execution.
Roles & Responsibilities:

- **Requirements Analysis**: Reviewed system requirements and authored test cases for both Model-in-the-Loop (MIL) and Hardware-in-the-Loop (HIL) validation.
- **Model Integration**: Integrated plant and controller models for MIL and HIL environments to ensure accurate simulation of motor behavior.
- **HIL Testing**: Conducted HIL testing using OPAL-RT platform to validate motor control features under real-time conditions.
- **MIL Testing**: Executed MIL testing with MATLAB and Simulink Test to verify control algorithms and system stability.
- **Automation**: Developed automated test scripts using Robot Framework and Python in PyCharm IDE, reducing manual effort and improving test efficiency.